SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1596

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 15
Duration : 60 Minutes

Code of Conduct
I **NIVASINI VM 192511356** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: **Nivasini VM**

Assessment Tasks

MCQ Test

1. **A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?**

   a) 5 TB
   b) 10 TB
   c) 15 TB
   d) 20 TB

Answer: c) 15 TB
5 TB × 3 regions = **15 TB**

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

~4 minutes
b) ~40 minutes
c) ~4 hours
d) ~24 hours

Answer:a) ~4 minutes
99.99% uptime means 0.01% downtime ≈ **4.3 minutes/month**

3.A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?

a)5000
b) 6000
c) 7000
d) 10,000

Answer: b) 6000
60,000 ÷ 10 = 6000 requests/server/second

4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?

1 hour
b) 10 hours

c) 1.1 hours
d) 5.5 hours

Answer: c) 1.1 hours
500 GB × 8 = 4000 Gb
4000 ÷ 1 Gbps = 4000 seconds ≈ 1.1 hours

5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

a) G~4 hours
b) ~8 hours
c) ~20 hours
d) ~40 hours

Answer:b) ~4 hours
99.95% availability → 0.05% downtime
8760 × 0.0005 = 4.38 hours/year

6. A cloud provider offers pay-as-you-go pricing at $0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?

a) $50
b) $500
c) $1000
d) $2000

Answer:b) $500
10 TB = 10,000 GB
10,000 × $0.05 = $500

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

a) 9.8 Gbps
b) 9.6 Gbps
c) 8.0 Gbps
d) 7.5 Gbps

Answer:a) 9.8 Gbps
10 Gbps × (100% − 2%) = 9.8 Gbps

8.A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

a) 91.25 GB
b) 85 GB
c)100 GB
d) 120 GB

Answer:a) 91.25 GB
250 MB × 365 = 91,250 MB = 91.25 GB

9.A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

a) 500 Mbps
b) 250 Mbps
c) 200 Mbps
d) 1000 Mbps

Answer:a) 500 Mbps
5 Mbps × 100 users = 500 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

a) 262 minutes
b) 365 minutes
c) 525 minutes
d) 150 minutes

Answer:a) 262 minutes

1 year = 365 × 24 × 60 = 525,600 minutes

- Downtime = 0.05% of 525,600
- = 525,600 × 0.0005
- = 262.8 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

a) 4,000
b) 8,000
c) 6,000
d) 10,000

Answer:b) 8,000
200 physical servers × 40 VMs per server
= 8,000 VMs

12. **A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?**
    a. 4 GB
    b) 6 GB
    c) -8 GB
    d) 10 GB

Answer:c) -8 GB

13. **A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?**
    a. 16
    b) 32
    c) 64
    d) 128

Answer:b) 32

Calculation:
64 cores × 2 = 128 vCPUs
128 ÷ 4 vCPUs per VM = 32 VMs

14. **A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?**
    a. 40 kW
    b) 50 kW
    c) 60 kW
    d) 70 kW

Answer: a) 40 kW

Calculation:
Power reduction = 60% of 100 kW
= 100 × 0.60 = 60 kW

New power = 100 − 60 = 40 kW

15. **A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?**

a. 6,400
b) 12,800
c) 25,600
d) 50,000

Answer:c) 25,600

Calculation:
100 servers × 64 vCPUs = 6,400 vCPUs
6,400 × 4 = 25,600 vCPUs

16. **Which cloud deployment model is used for organizations that require both private and public cloud resources?**

A): Hybrid Cloud
B): Public Cloud
C): Private Cloud
D): Community Cloud

Answer:A): Hybrid Cloud

17.Which cloud security strategy ensures that sensitive data remains private even if intercepted?

A): Encryption
B): Load balancing
C): Virtualization
D): Resource pooling

Answer:A) Encryption

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

a: 5000 GB
b: 7000 GB
c: 6000 GB
d: 7500 GB

Answer:b) 7,000 GB

Calculation:
50 VMs × 20 GB/day = 1,000 GB/day
1,000 × 7 days = 7,000 GB/week

19.A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

a: 56 GHz
b: 58 GHz
c: 64 GHz
d: 70 GHz

Answer:a) 56 GHz

Calculation:
Total processing power = 32 × 2.5 GHz
= 80 GHz

Utilized power = 80 × 70%
= 56 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

a: 1.5 sec
b: 2 sec
c: 1 sec
d: 3 sec
Answer: a) 1.5 sec

Calculation:
Total delay per request = 5 ms + 10 ms = **15 ms**
For 100 requests = 15 × 100 = **1500 ms**
1500 ms = **1.5 sec**

21.If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

a: 600 GB
b: 500 GB
c: 700 GB
d: 800 GB

Answer:a) 600 GB

Calculation:
100 VMs × 200 MB/day = 20,000 MB/day
20,000 × 30 = 600,000 MB
600,000 MB = 600 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

a: 21.6 min
b: 36 min
c: 43.2 min
d: 18 min
Answer:a) 21.6 min

Calculation:
30 days = 30 × 24 × 60 = 43,200 minutes
Downtime = 0.05% × 43,200
= 43,200 × 0.0005
= 21.6 minutes

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

a: 120 MB
b: 100 MB
c: 150 MB
d: 200 MB
Answer:a) 120 MB

Calculation:
200 requests/sec × 10 KB = 2,000 KB/sec

2,000 × 60 = 120,000 KB/min
120,000 KB ≈ 120 MB/min

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.
a: 180
b: 150
c: 120
d: 160
Answer:a) 180

Calculation:
75% of 120 VMs = 120 × 0.75 = 90 VMs
90 × 2 vCPUs = 180 vCPUs

25.A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?

a: 1200
b: 1000
c: 800
d: 600
Answer:a) 1200

Calculation:
20 racks × 20 servers = 400 servers
400 × 3 vNICs = 1,200 vNICs

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

a: 12 min
b: 10 min
c: 6 min
d: 8 min
Answer:a) 12 min

Calculation:
60 VMs ÷ 10 VMs per batch = 6 batches

Each batch takes 2 minutes.
$6 \times 2 = 12$ minutes

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.
Formula:
Energy = Units × Power × Time (hrs)
a: 420 kWh
b: 480 kWh
c: 400 kWh
d: 500 kWh

Answer:a) 420 kWh

Calculation:
Energy = Units × Power × Time
= 5 × 3.5 kW × 24 hours
= 420 kWh/day

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?

a: 32 TB
b: 35 TB
c: 30 TB
d: 28 TB
Answer:a) 32 TB

Calculation:
20 servers × 2 TB = 40 TB
20% redundancy = 40 × 0.20 = 8 TB
Usable capacity = 40 − 8 = 32 TB

***29.*Which of the following is NOT a benefit of virtualization?**

i. Reduced hardware costs
b) Increased energy consumption
c) Improved resource management
d) Faster deployment of applications

Answer:b) Increased energy consumption

30.**Which feature in virtualization allows multiple OS environments to run simultaneously?**

i. Hypervisor
   b) Load balancer
   c) Firewall
   d) BIOS

Answer:a)Hypervisor